

Final Project (Group/Individual project)- Due date-May 12, 11:59 pm

***** No Late submissions will be accepted.**

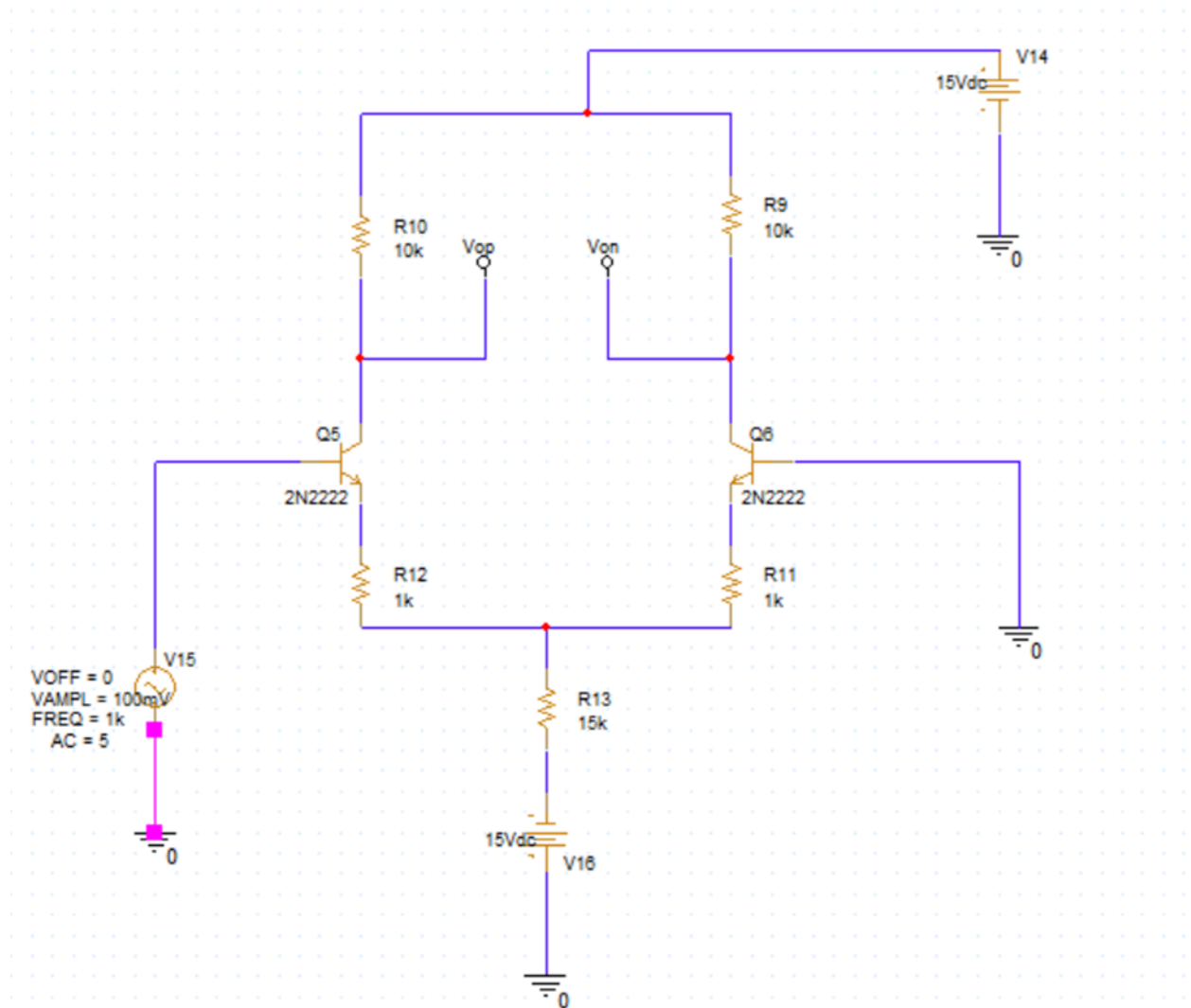
***** A maximum of two people can be in a group**

Objectives of the final project:

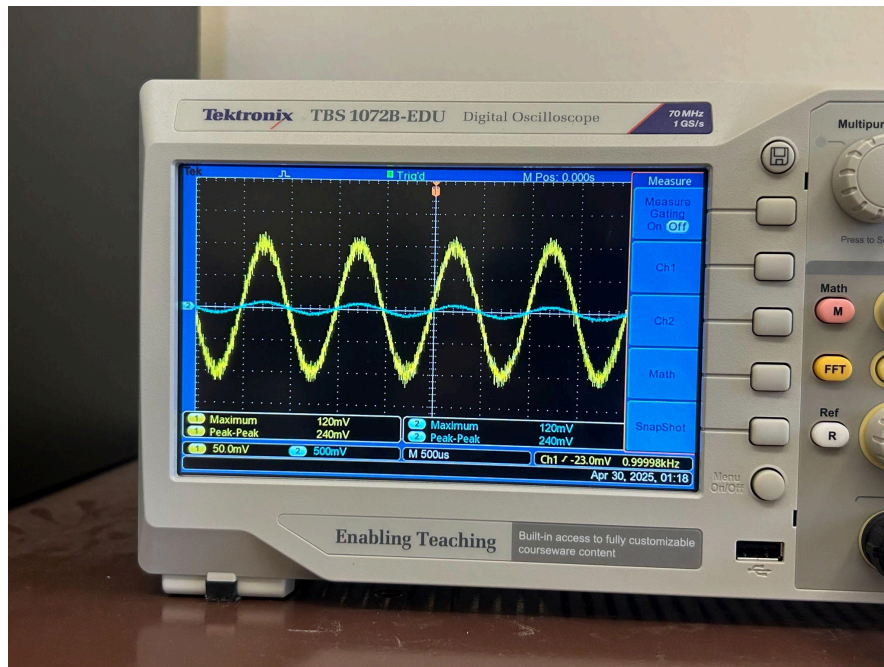
1. Explore previous lab concepts to build a differential and multistage amplifier.
2. Write a good technical report and demonstrate your learning and observations.

Project Description: Experiment #1 (Differential Amplifier):

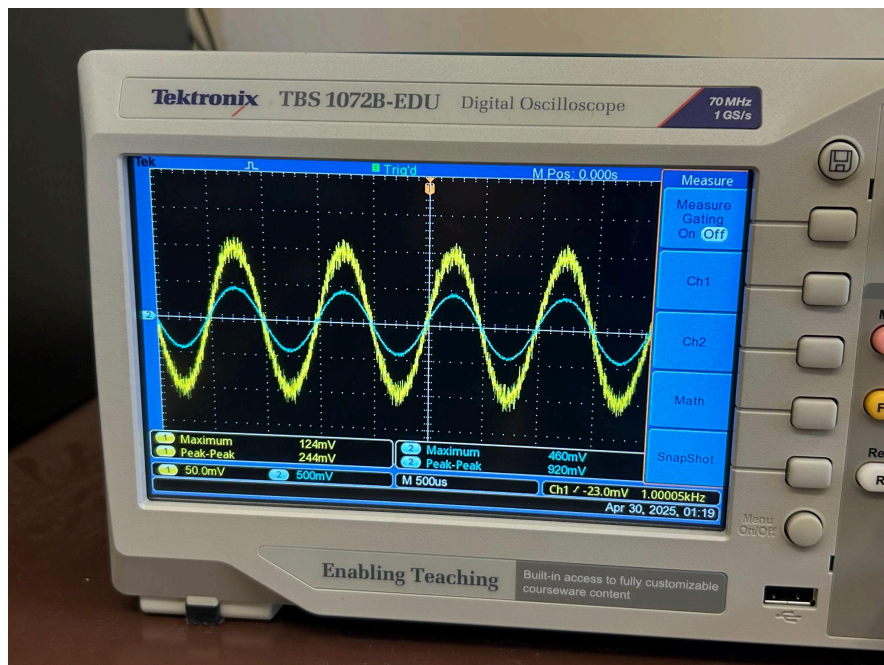
Suppose you are working as an audio engineer and trying to reduce the SNR. Design a BJT differential amplifier as shown below to reduce the SNR. Determine the voltage gain. Record the input and output waveforms. Compare your simulated and measured results and analyze your observations.

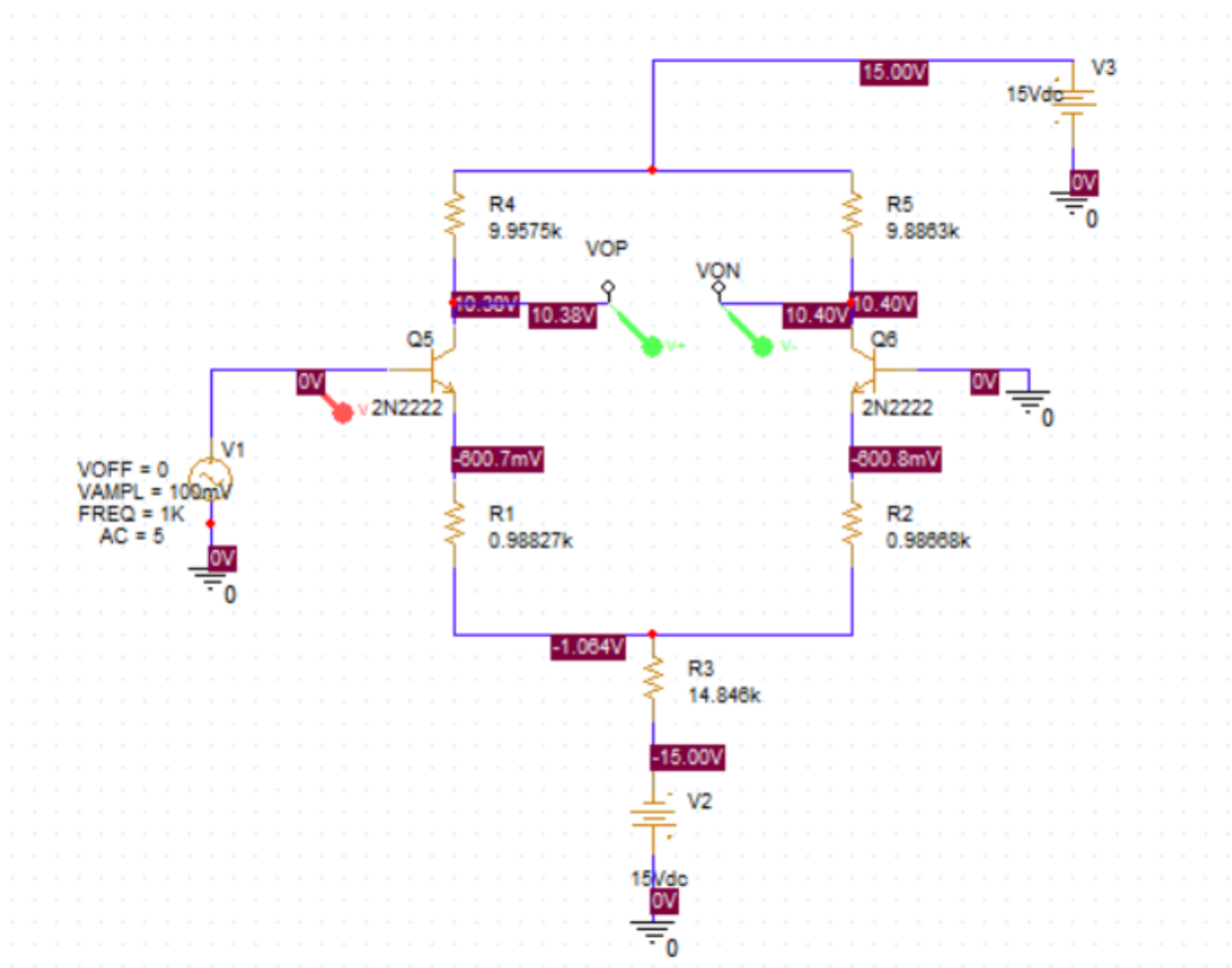
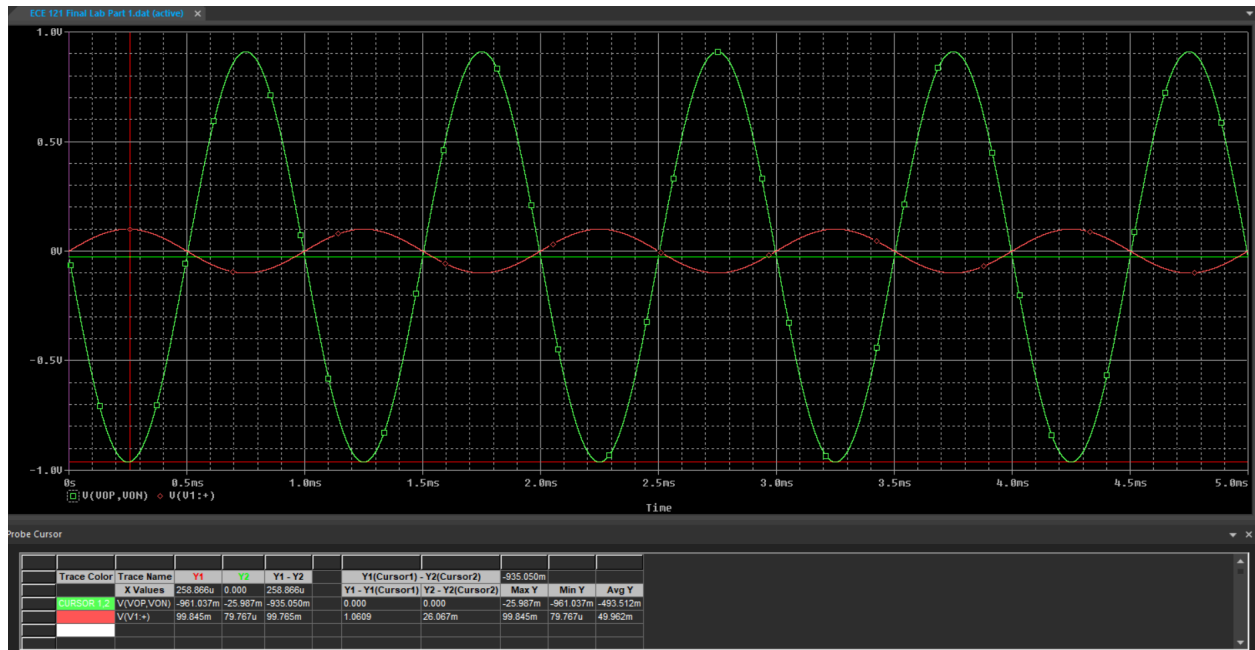


2X 2N3904
 R9=9.8863Kohm
 R10=9.9575Kohm
 R11=0.98668Kohm
 R12=0.98827Kohm
 R13=14.846Kohm
 Vin(Yellow) vs Vop(Blue)



Vin(Yellow) vs Von(Blue)

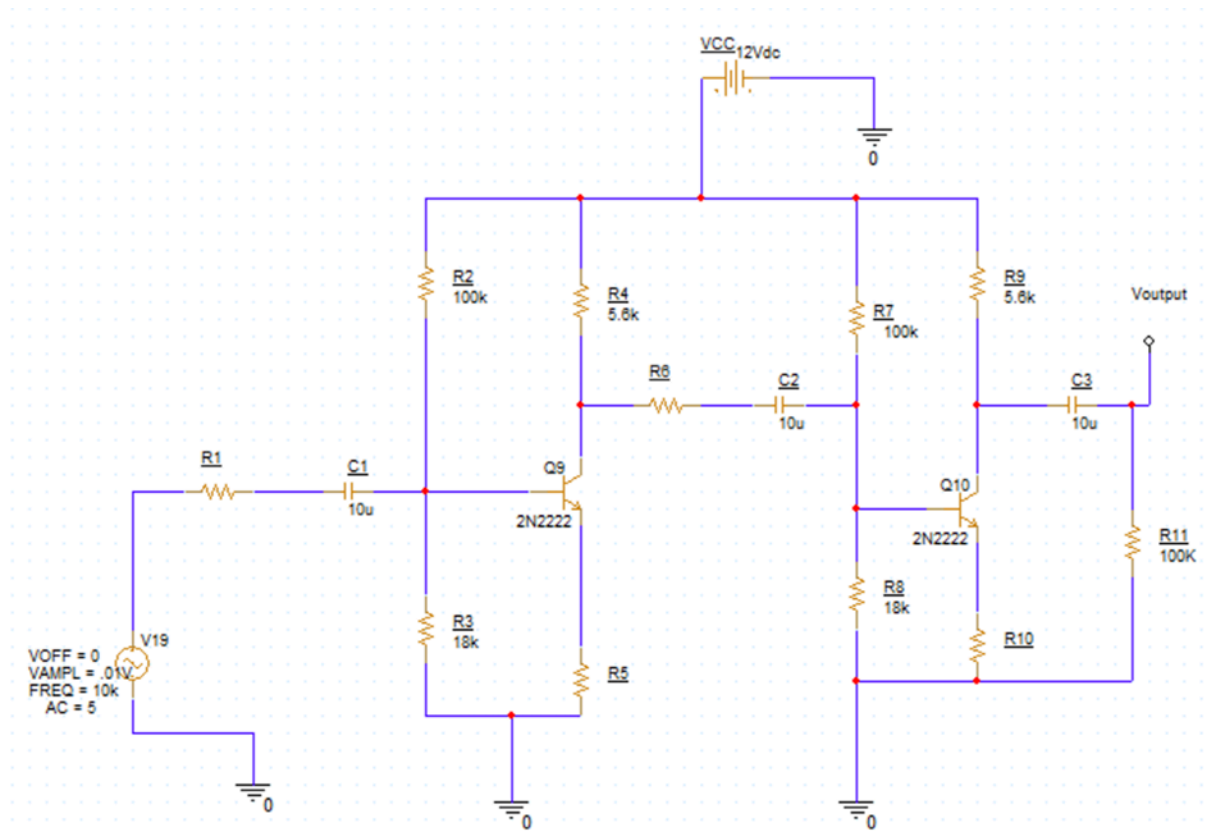


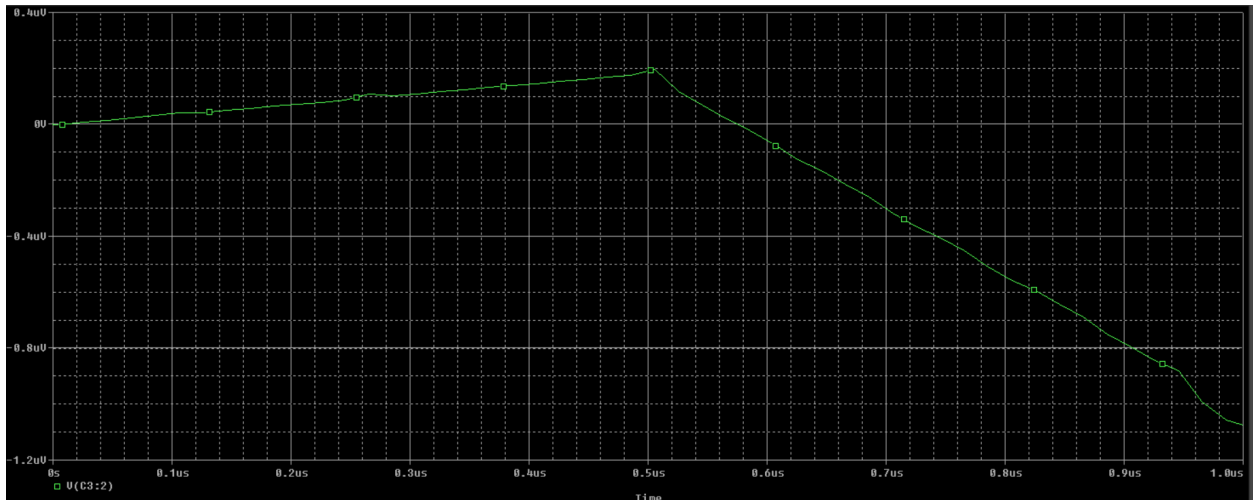
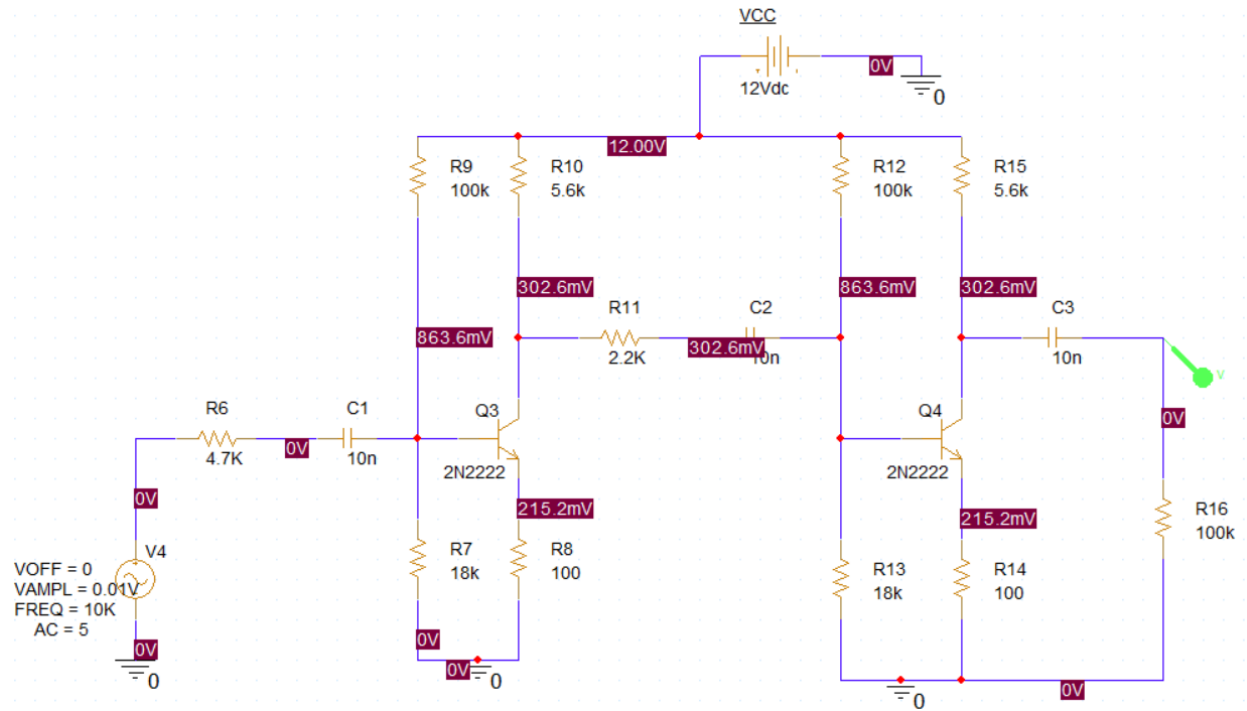


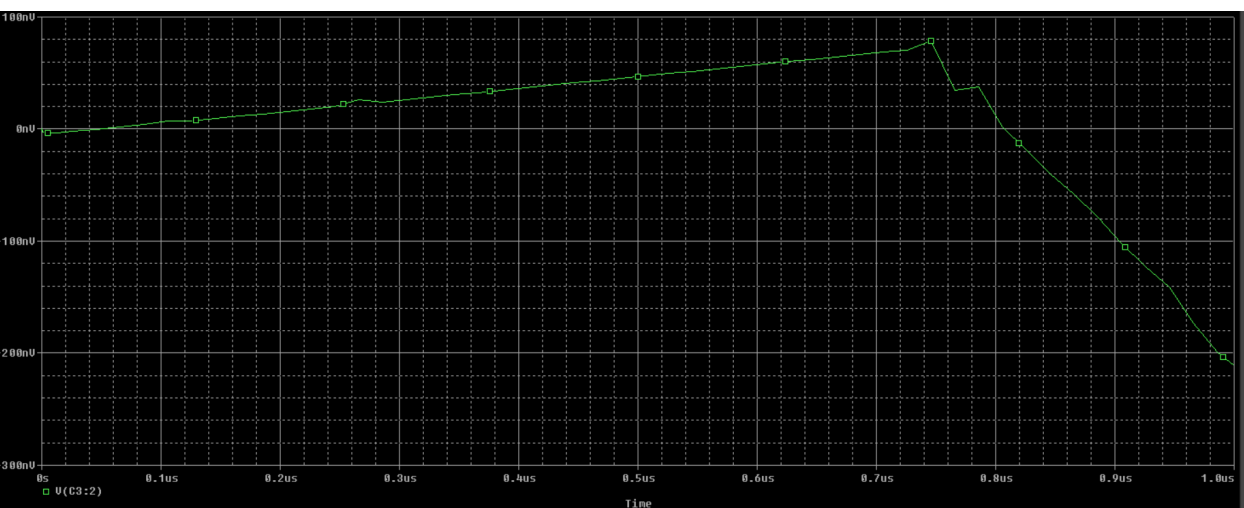
Conclusion: In my measured results I got 0.680mV as the output voltage and in the simulation I got an output of 0.920mV. I think the difference in results could be lab equipment error not being the same as the simulation however the wave forms looked the same.

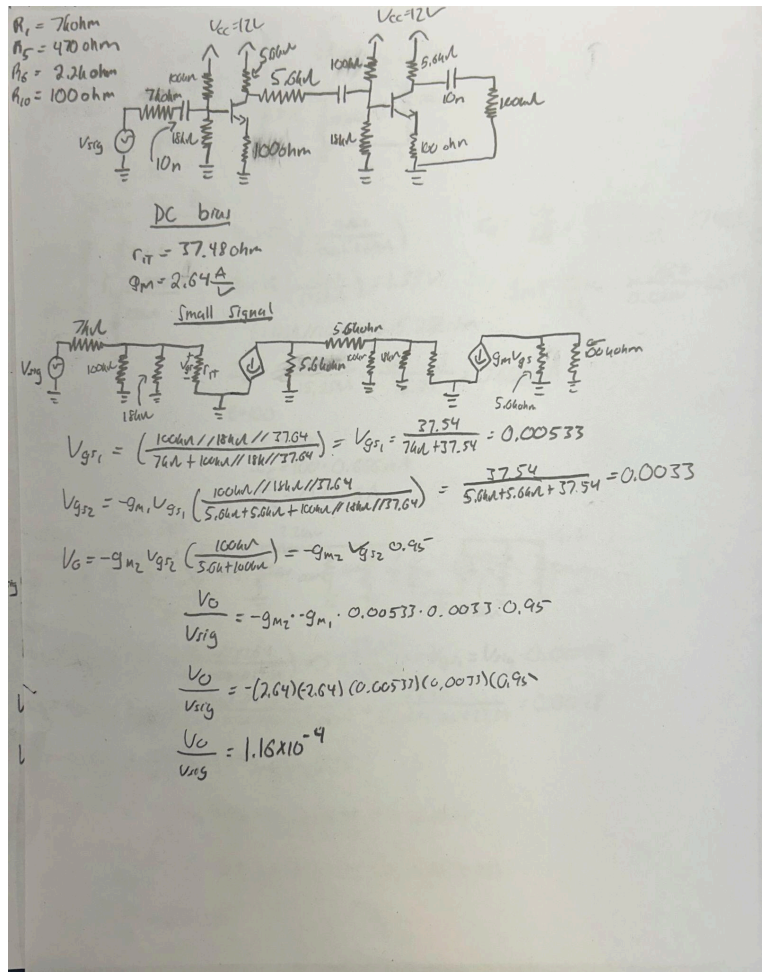
Project Description Experiment #2 (Multistage Amplifier):

Suppose you are working as an audio engineer and trying to amplify the input audio signal. Design an Audio amplifier using a multistage amplifier as shown below. Adjust the value of R1, R6, R5 and R10 to amplify the input signal. What resistance values do you use, and what is the gain in this case? Why did you choose this value? Briefly explain through calculations. Record the input and output simulated waveforms.









$R_1 = 7\text{K}\Omega$

$R_5 = 100\text{ }\Omega$

$R_6 = 5.6\text{K}\Omega$

$R_{10} = 100\text{ }\Omega$

Conclusion: After going through many different resistor combinations I found that if you kept the R_5 and R_{10} constant for both and increased the resistance of R_1 and R_6 it would decrease the gain which matches my calculations and simulations. Since R_5 and R_{10} control the gain of each stage so I kept them constant. In the second example I made the R_1 and R_6 resistors greater in resistance so the gain would be less than the first example.

Grade Details: The project reports will be based on the following scheme:

Objectives and background	10%
Calculation	20%
Result & Analysis (Simulation)	25%
Completion of work	15%
Documentation & Report writing	30%

Objectives and background	Member 'Frank Tamburro'	Member 'B'
Calculation	X	
Result & Analysis (Simulation)	X	
Completion of work	X	
Documentation & Report writing	X	